Reinforcement Learning for Adaptive Quantitative MRI in a Validated Digital Phantom

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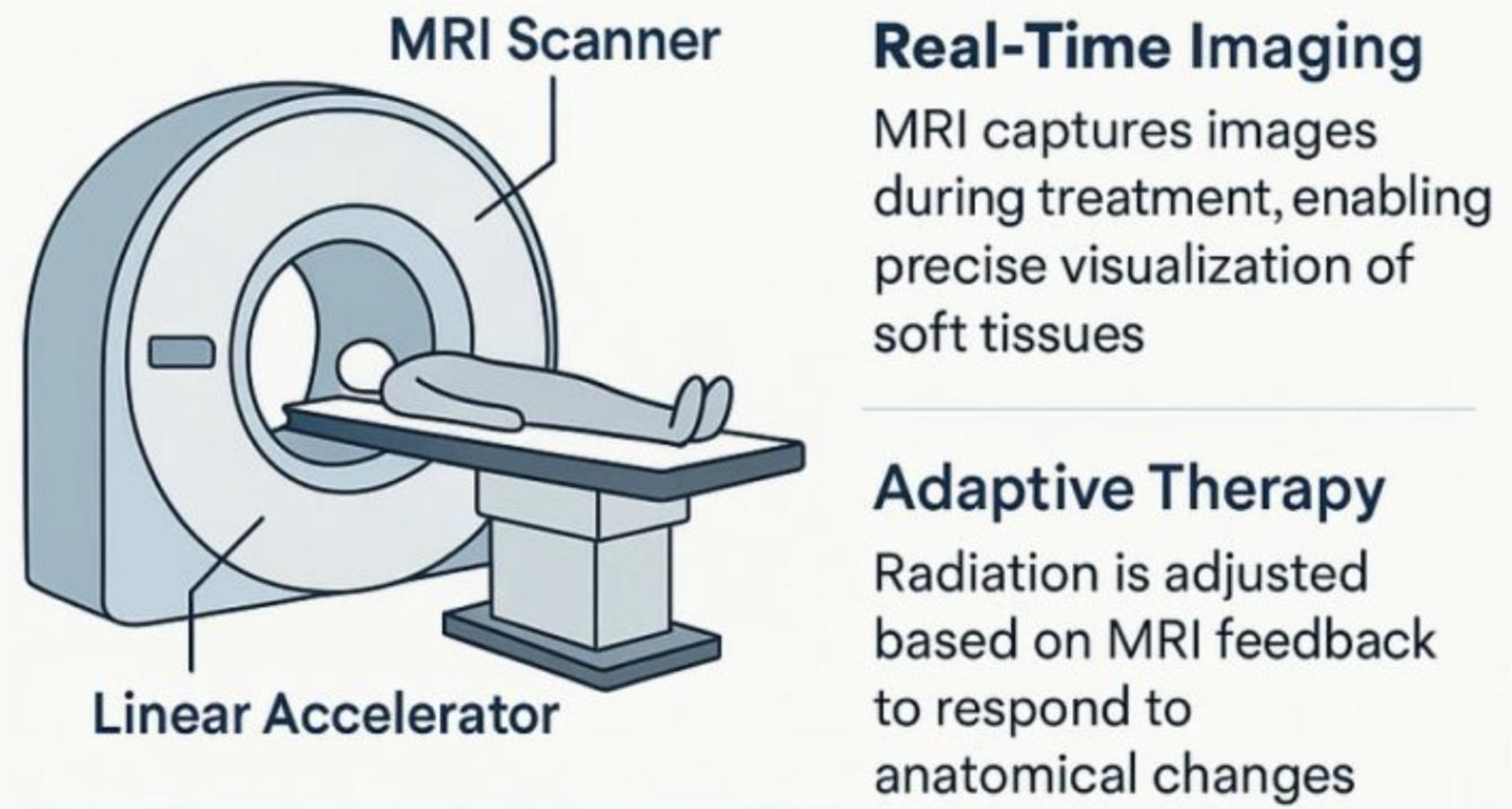
Does an agent that uses its current tissue estimates to choose the next acquisition outperform a static sequence?

Agenda

- What is Adaptive qMRI?
- Existing research
- Building a digital system phantom (MRISystemPhantom.jl)
- Validating the Simulator (KomaMRI)
- Multi-fidelity training (making RL training cheap)
- RL experiments
- Results
- Future work & Limitations
- Conclusion/Impact of this work

MR-Linac is a combination of an MRI scanner and a Linear Accelerator (radiation emitter)



Improved tumour delineation leads to more focused radiation and better patient outcomes.

However, scan time is a major bottleneck which is the problem this research is solving.

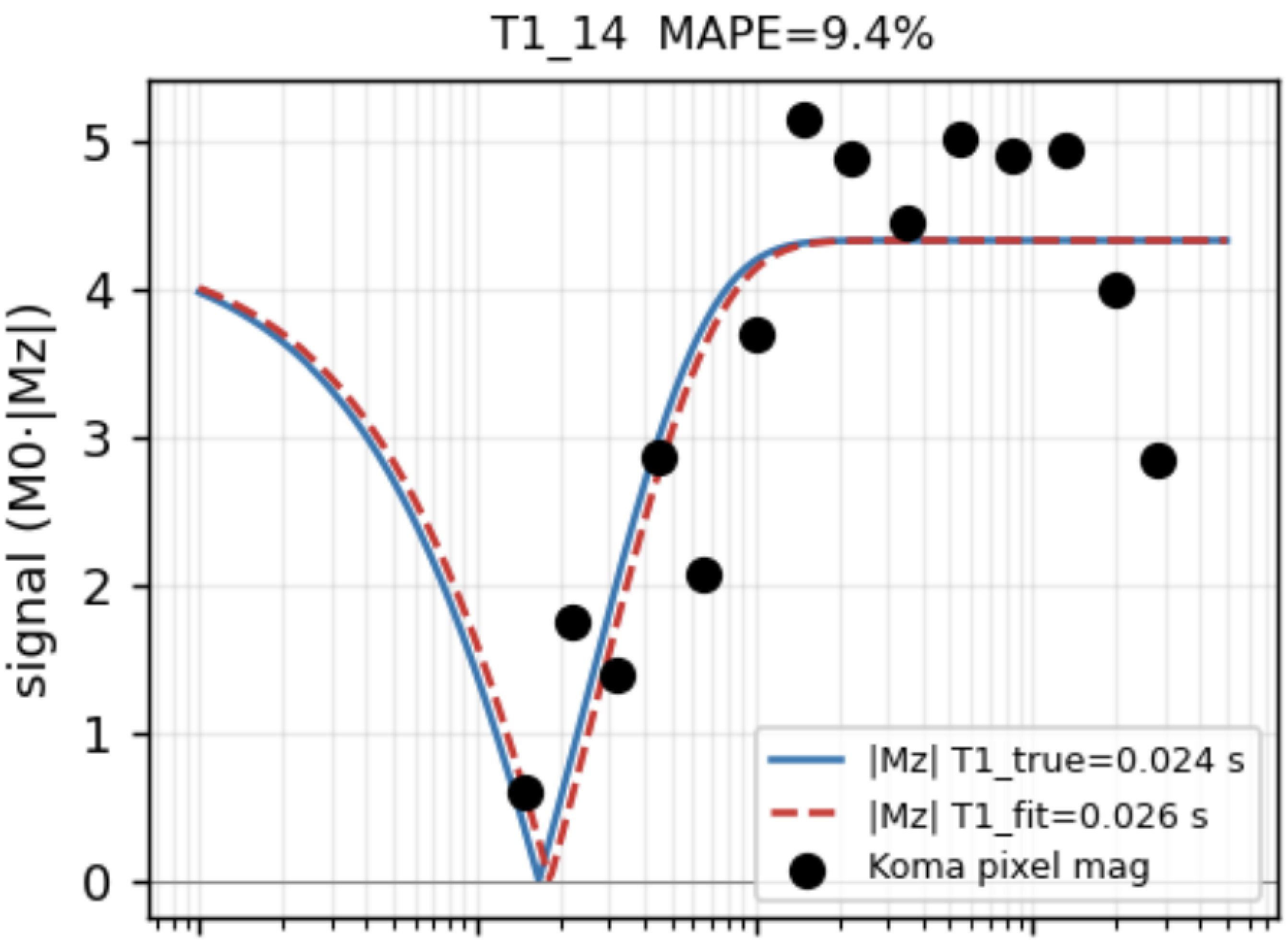
Significant ML research has been conducted to reduce acquisition time in traditional MRI scans

RL / Learned Control of MRI	Method	Objective	Quantitative?
Walker-Samuel et al. (2023)	Deep RL (DDPG)	Shape Classification	✗
Pineda et al. (2020)	Deep RL (DQN)	K-Space Recon Quality	✗
AUTOSEQ (2018)	Bayesian Optimisation	Image MSE (1-D Toy)	✗
DeepRF (2021)[2]	Deep RL	RF Pulse Waveform Design	✗
THIS WORK	**Deep RL (PPO)**	**Fitted-T1 Error**	✓

[2] Shin et al., DeepRF, Nat. Mach. Intell. 2021.

Quantative MRI provides better tissue differentiation using T1/T2 values

- T1/T2 is the rate of longitudinal/transverse magnetization relaxation
- Measuring T1/T2 requires multiple images at different timings to fit the relaxation curve
- Inversion time (TI) is the main acquisition parameter you can change for T1 fitting
- This increases acquisition time – so can't be used on the MR-Linac
- There are more informative acquisitions than others dependent on the T1/T2 values being measured
- Adaptive qMRI: to reduce scan time – we should adapt our acquisition parameters based on the live T1/T2 estimate



Existing Adaptive qMRI research has been limited to single T1/T2 estimates

Adaptive qMRI exists — but never a learned multi-voxel policy

Quantitative MRI Sequence Design	Method	Adaptive?	Multi-voxel?
Beracha et al. (2023)	Bayesian Posterior + CRLB Lookup	✓	✗
MRF Schedule Optimisation	Differentiable, Offline	✗	✓
MIMOSA (2026)[1]	Optimised Multi-Echo, Offline	✗	✓
THIS WORK	**Learned RL Policy (PPO)**	✓	✓

[1] Chen et al., MIMOSA, Magn. Reson. Med. 2026.

In 2023 Beracha delivered 2-4x acceleration vs. static sequences
However, it is limited to a single voxel (a 3D tissue pixel with one set of T1/T2 estimates)
This work generalizes to the multi-voxel case.
A harder problem so RL is used since no mathematical solution to minimize estimate variance (CRLB sequence)

Two key implementation challenges for RL

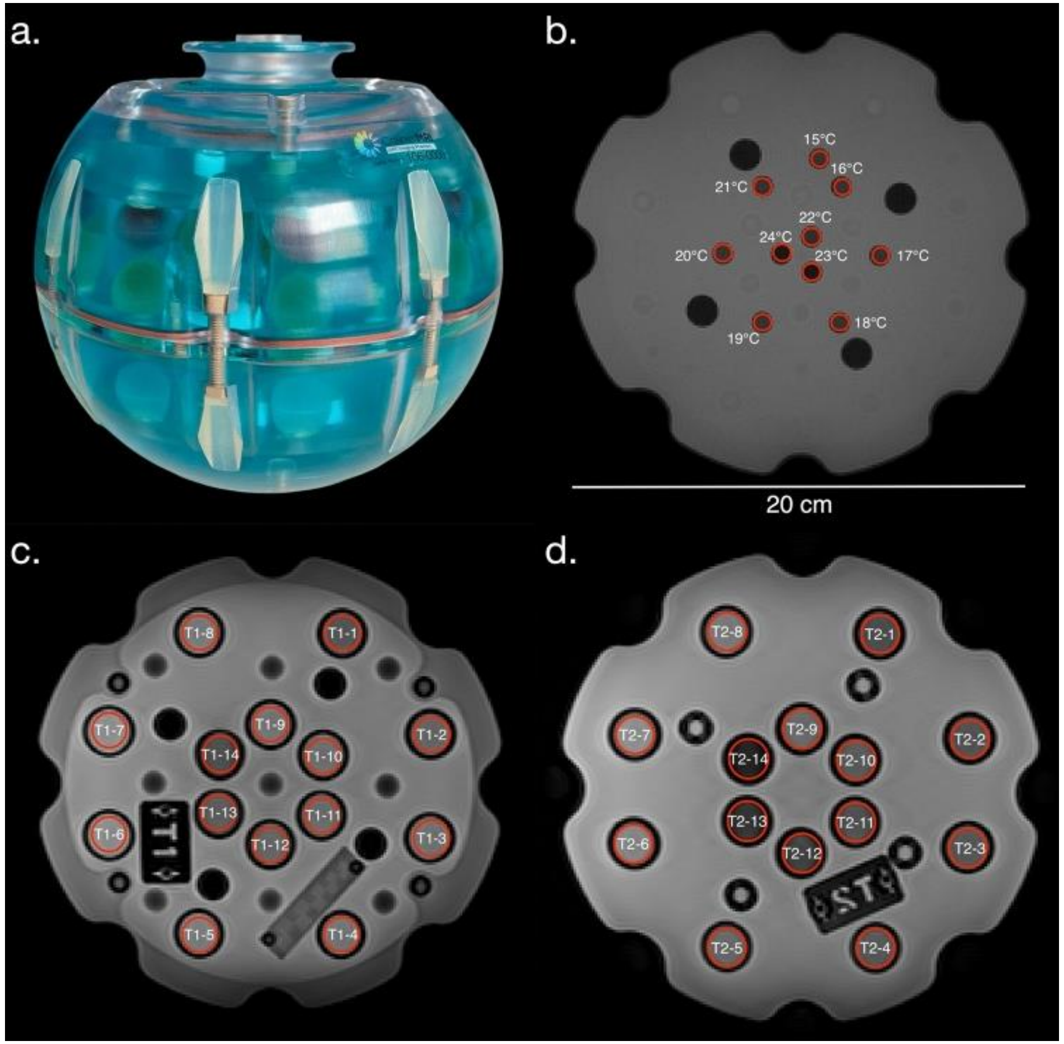
RL: where an agent learns by interacting with an environment and receiving rewards for its actions.

RL needs to simulate millions of MRI acquisitions leading to two challenges:

- C1 — Simulator + T1 fitting pipeline validation for qMRI using a digital phantom (object with known T1/T2 values used to test & calibrate MRI scanners)
- C2 — Computational efficiency: KomaMRI simulator is slow → multi-fidelity training

(A1 / C1) MRISystemPhantom: an open-source digital phantom library

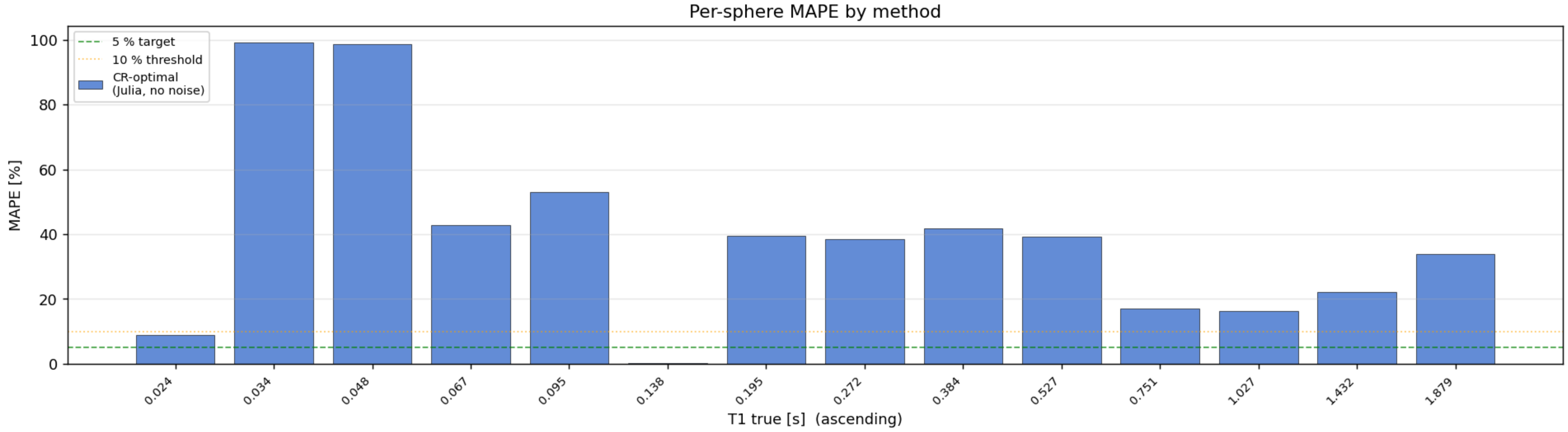
- QalibreMD Model 130 is the industry standard phantom
- Includes a plate of T1 calibration spheres that span the full clinical range (14 spheres, 24 ms–1.9 s)
- MRISystemPhantom.jl produces KomaMRI.Phantom objects
- Open-source
- Thoroughly tested & documented



(A2/C1) Uncovering KomaMRI failures using MRISystemPhantom's fitting pipeline

KomaMRI is a peer-reviewed, Julia MRI simulator

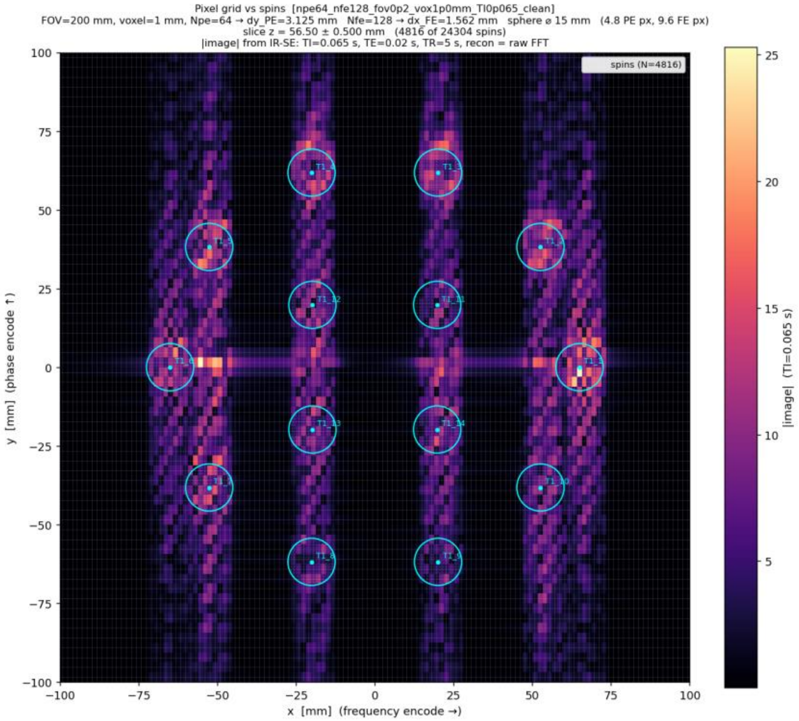
39.4% MAPE for a simple T1 recovery curve fit

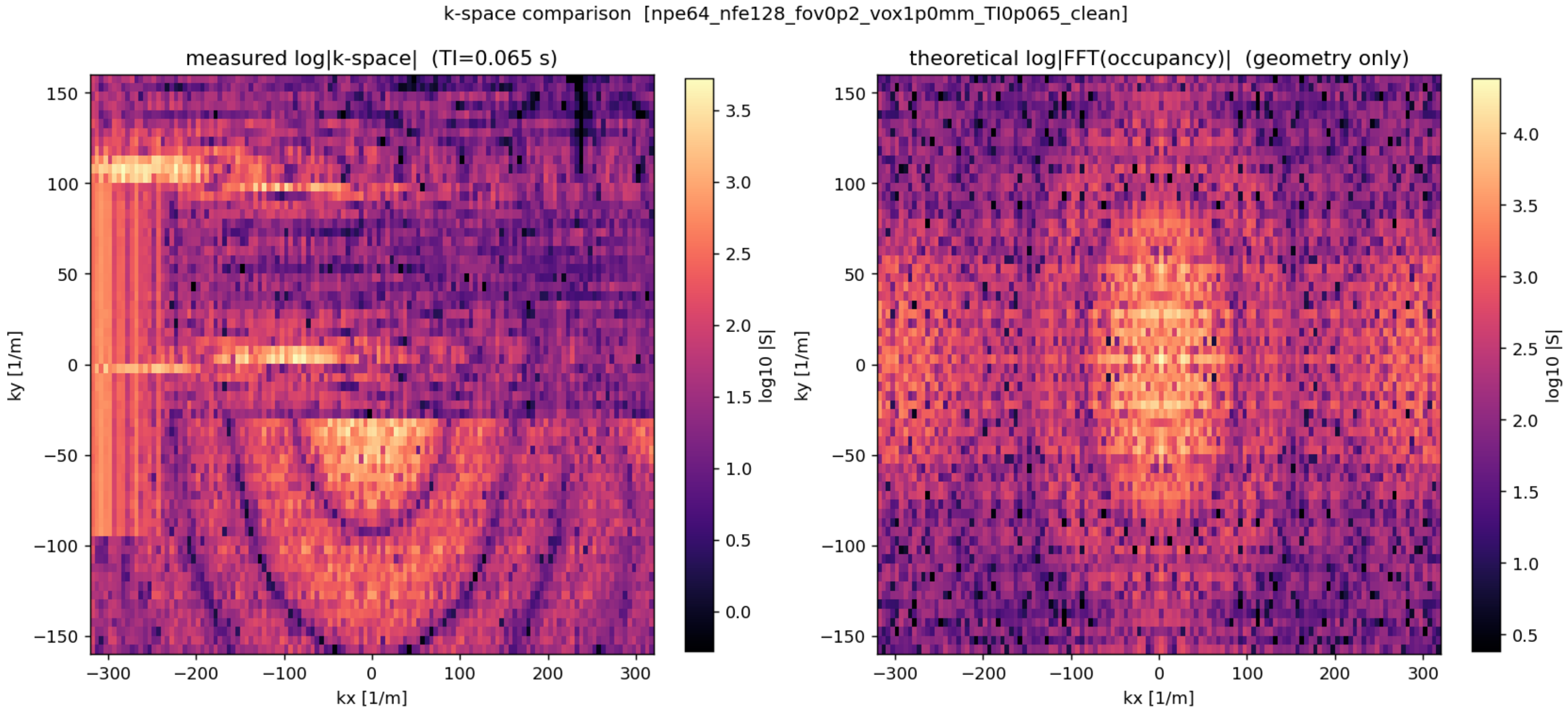


(A2/C1) Hard to differentiate physical phenomena from simulator inaccuracies

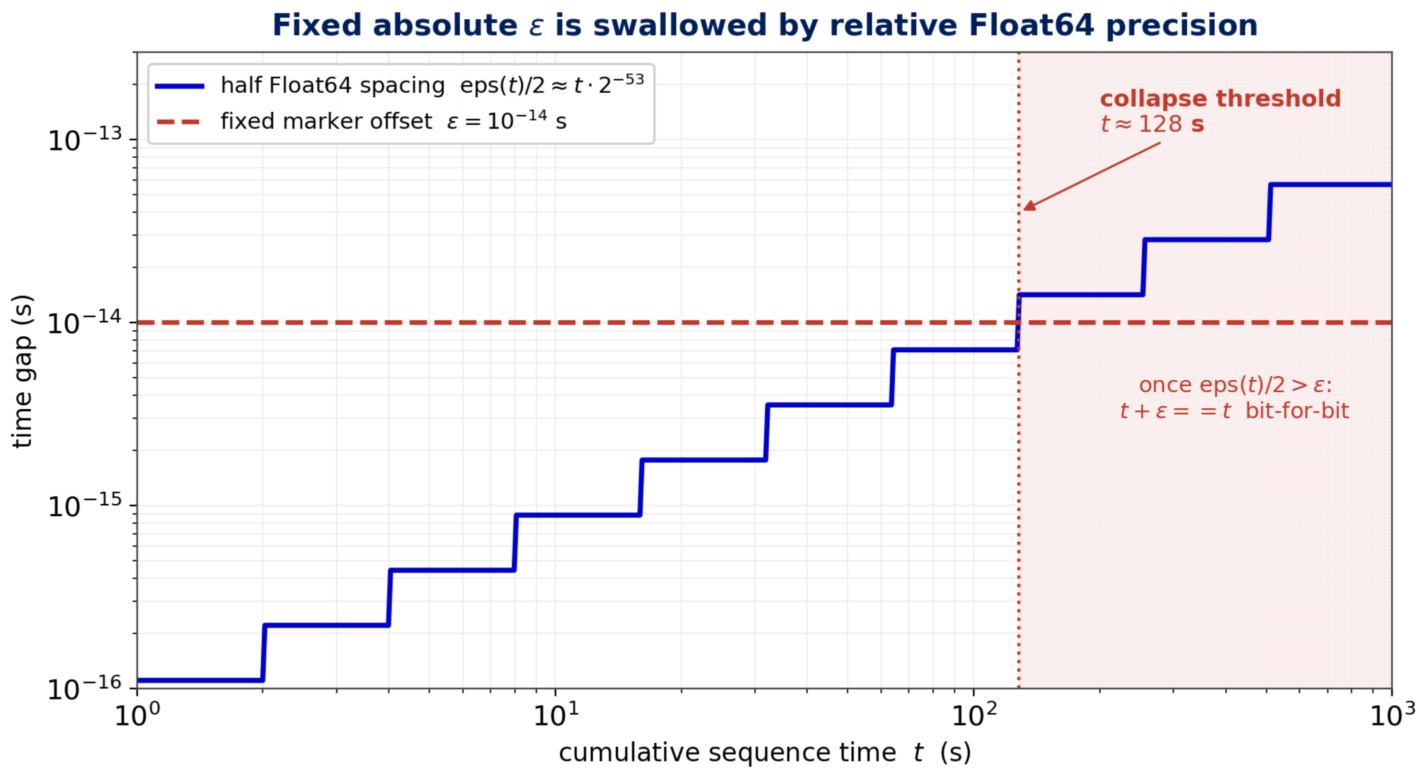
Too much distortion for Gibbs ringing

The giveaway: k-space collapses after a fixed cumulative time





(A2) Two floating-point bugs in KomaMRI, found & fixed



RF pulses in KomaMRI are modelled as trapeziums. The trapezium edges are a fixed ε = 1e-14 s

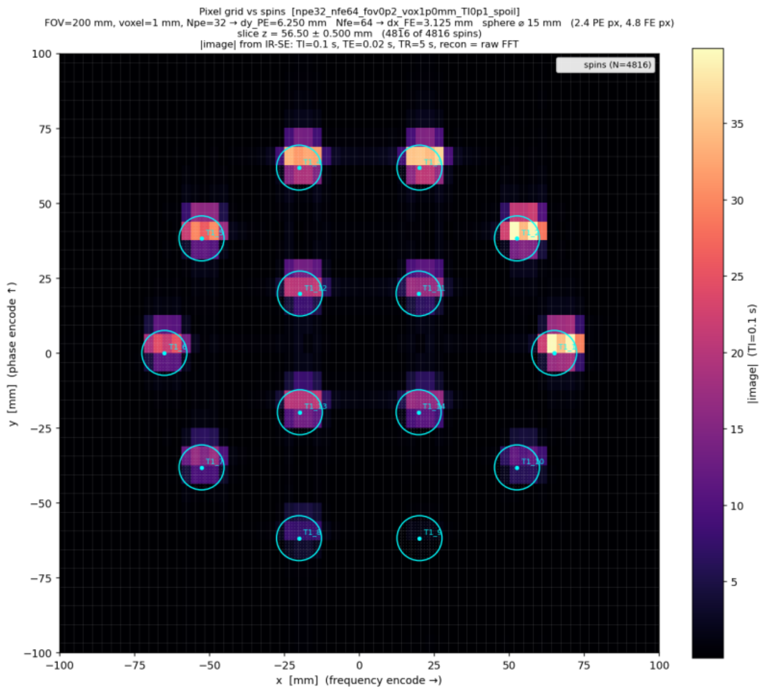
Float64 precision is relative: eps(t) ≈ t·2^{-52}. Therefore t+ε == t past ~128 s

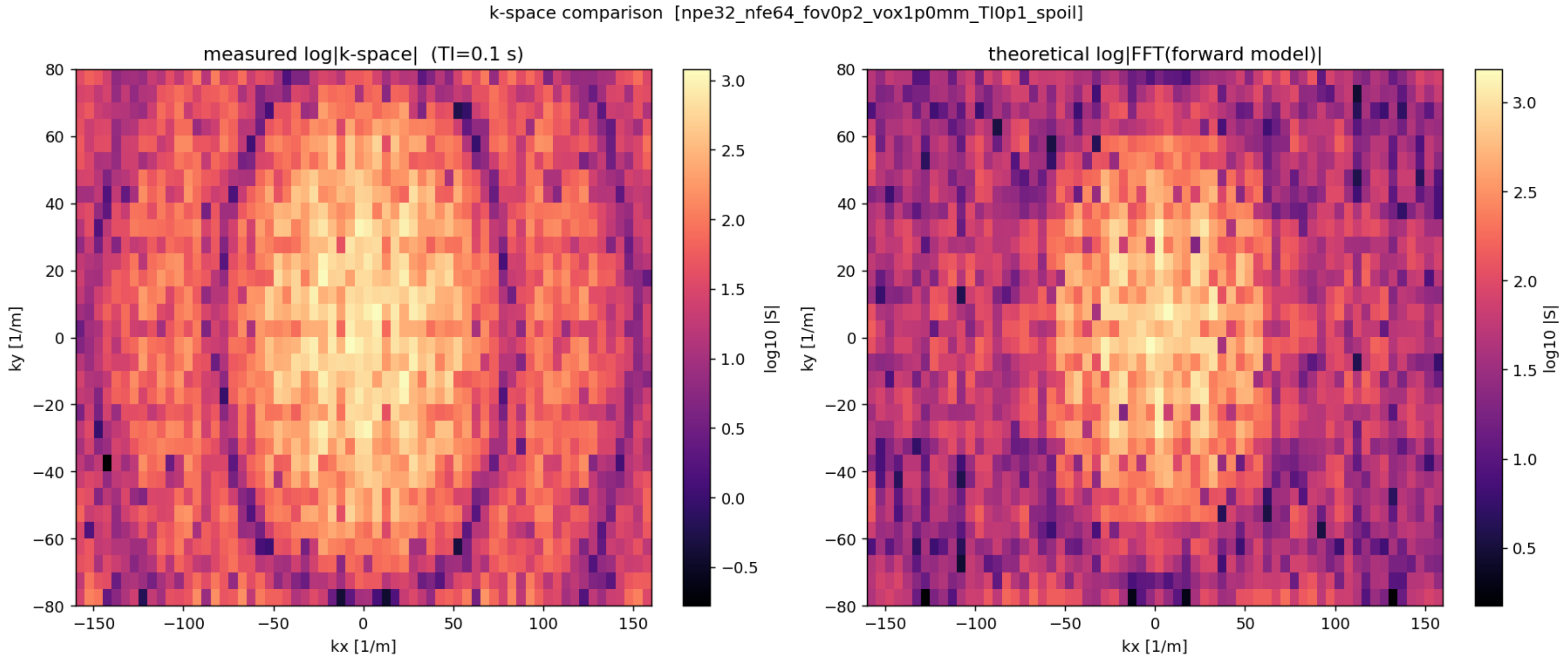
Fixed upstream (PR #780, #789)

(A2) Fully validated qMRI T1 fit

39.4 -> 0.48% MAPE for T1 fits

Simulator now quantitatively trustworthy for RL



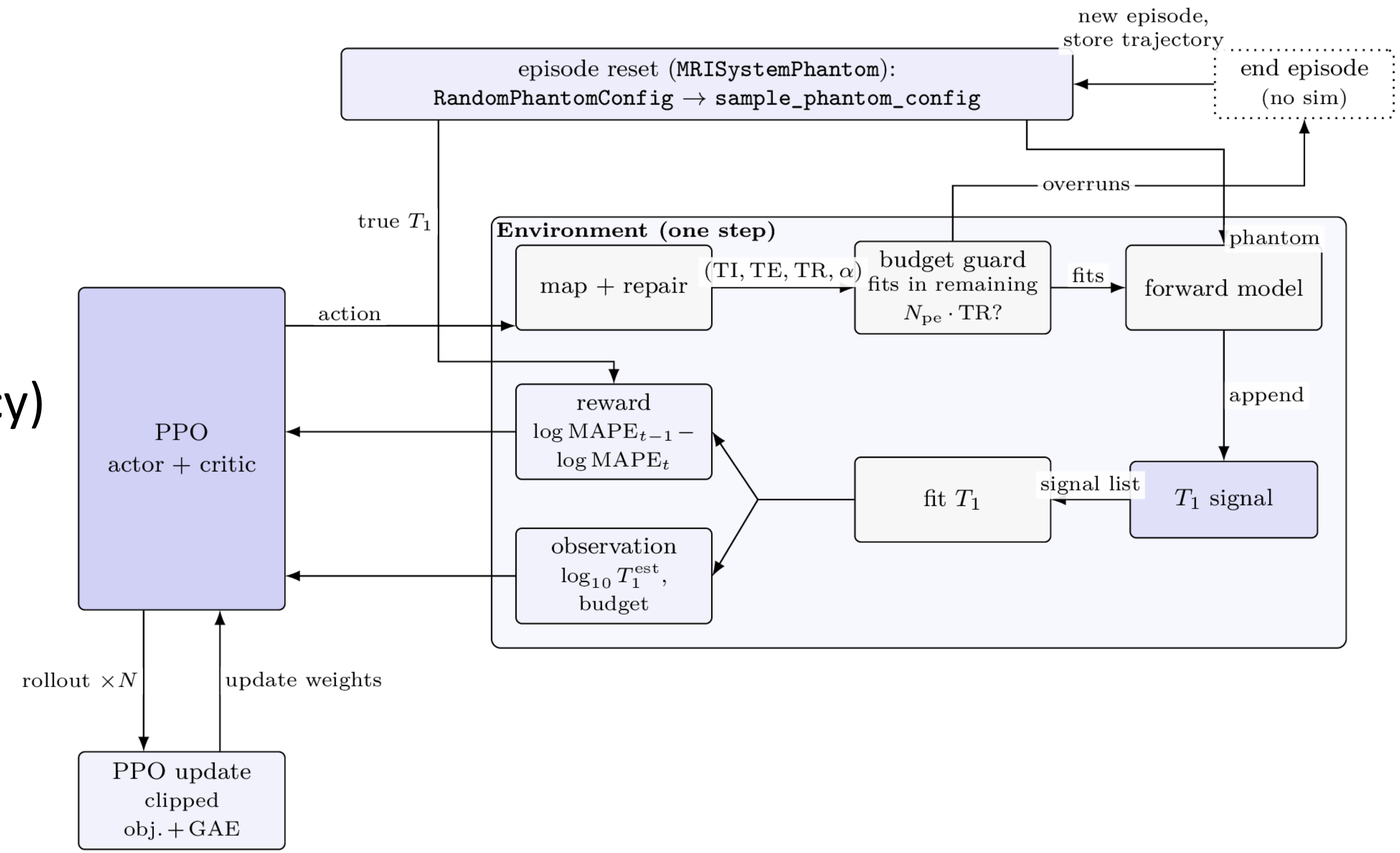


(A1/C1) Future work for MRISystemPhantom

- The current fitting algorithm uses a coarse grid search that limits T1 accuracy to ~0.5%
- Current fitting pipeline assumes perfect transverse spoiling between acquisitions
- Instead, switch to an EPG fitting or even end-to-end RL estimate
- Static sequences that ship with the library are limited (PULSEQ integration?)

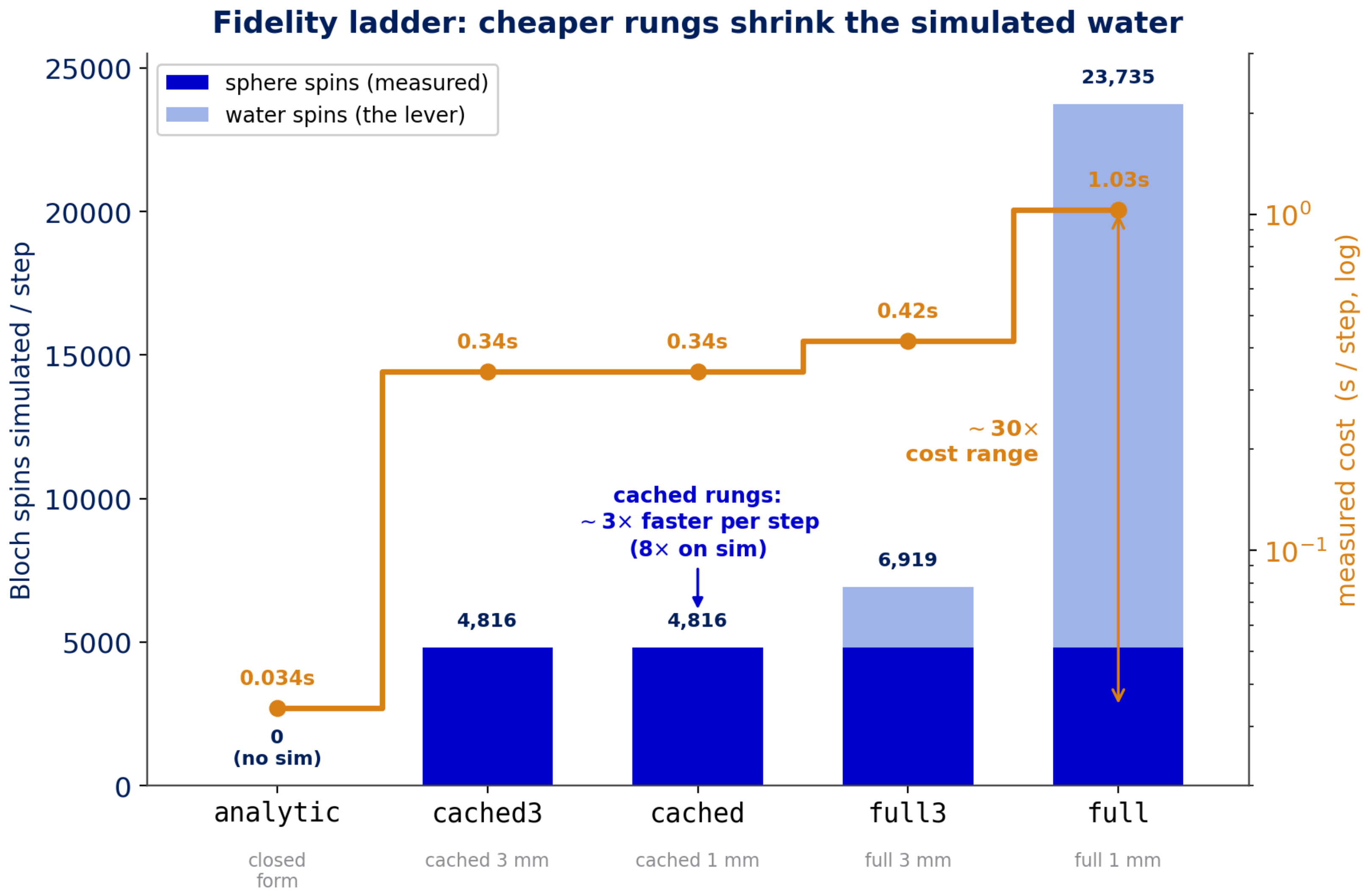
(A3) The RL experimental setup

- 14 sphere T1 fit, 240 secs
- State: per-sphere T1 est + budget
- Action: TI & TR
- Reward: Δlog MAPE
- RL algorithm: PPO (on-policy)
- σ=50 → SNR 17–28



A3 / C2 — The cost wall and the multi-fidelity ladder

- Full-Bloch step ≈ 4 s → 200k steps ≈ 9 days
- 80% of spins are water the agent never measures
- Lever: coarsen water · cache water



(A3 / C2) Cached water: exploiting the additive nature of simulated spins

$$S_{\text{full}} = S_{\text{spheres}} + S_{\text{water}}$$

re-simulate only the sphere spins each step

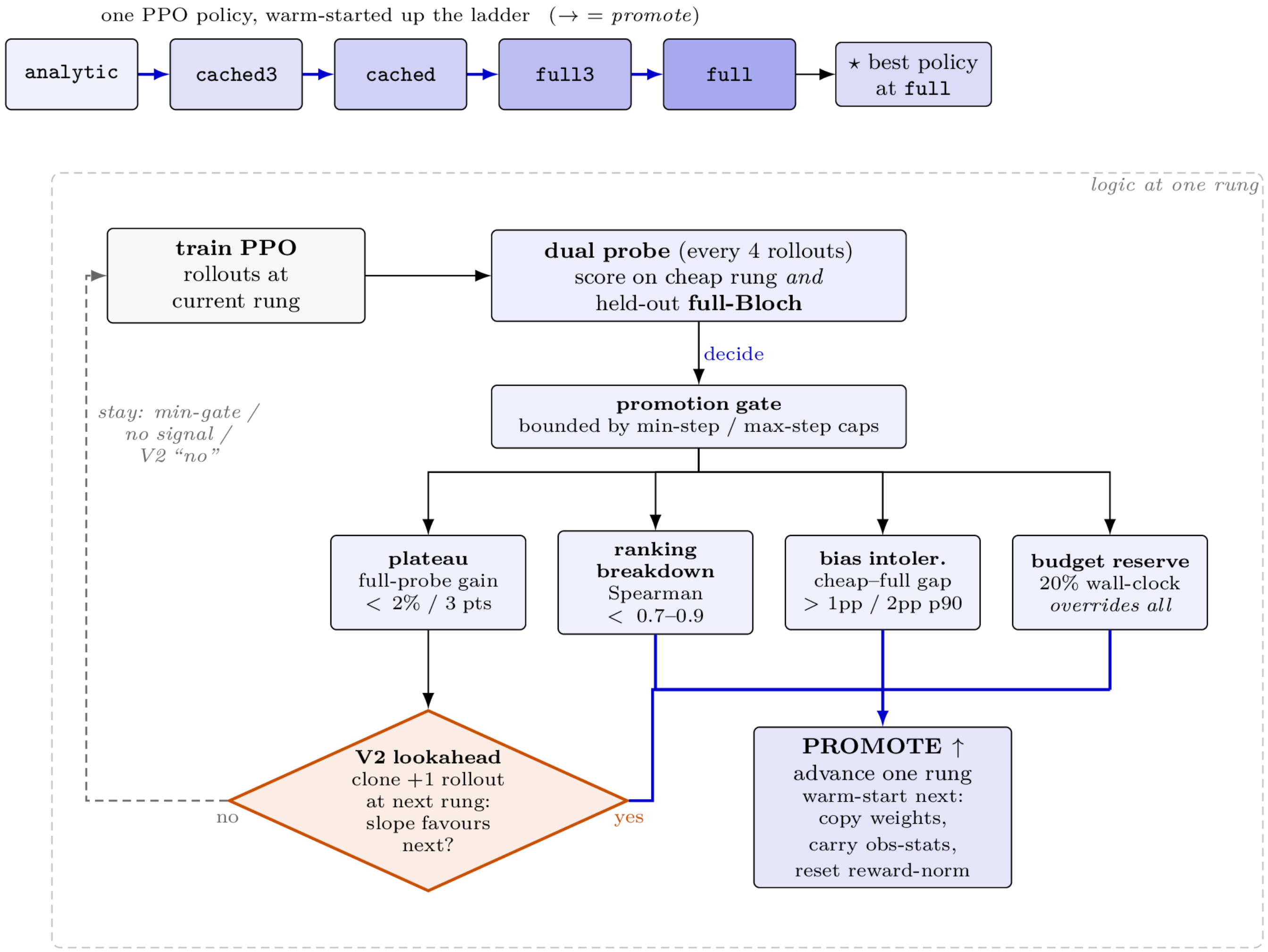
$$S_{\text{water}}[k] = M_z^{(k)}(\mathrm{TI}, \mathrm{TR}, \alpha)\sin\alpha \; W_\alpha[k]$$

homogeneous water factorises — only the scalar $M_z^{(k)}$ depends on the timing,
so the geometric template W_α is built once and reused

$\sim 8\times$ cheaper per step · matches full-Bloch T_1 fits to 0.12% (worst 1.15%)

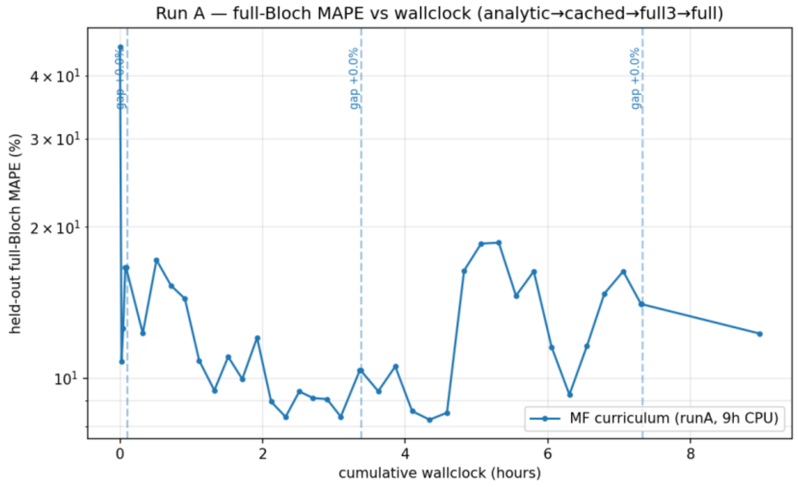
(A3) The trust problem: bias-aware promotion

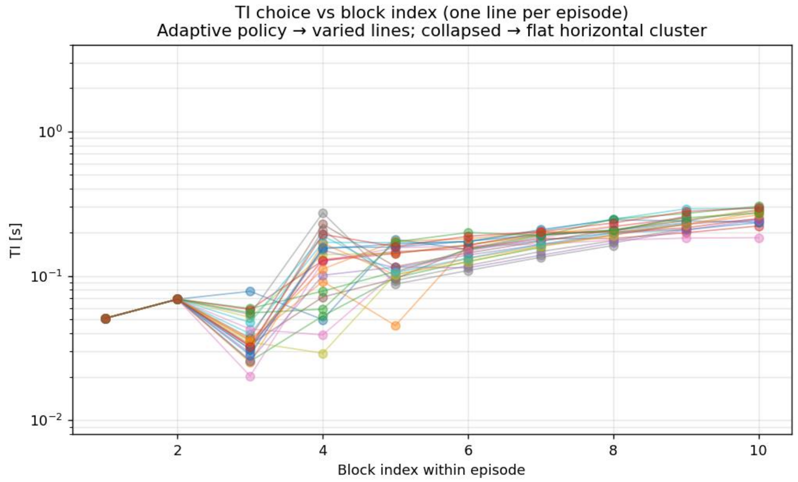
- Measure MAPE on the most accurate simulator every 4 rollouts (512 sims per rollout)
- If MAPE diverges with current simulator or training improvement stalls – promote to next stage



Adaptive RL agent 10.4% MAPE; CRLB sequence 4.7% MAPE (Run A)

- Agent is adaptive (TI tracks estimate) after initial probe
- Adaptivity not as useful. Too many, wide ranging T1 values
- Multi-fidelity training proved effective – used global best model (ended up coming from the cached layer)



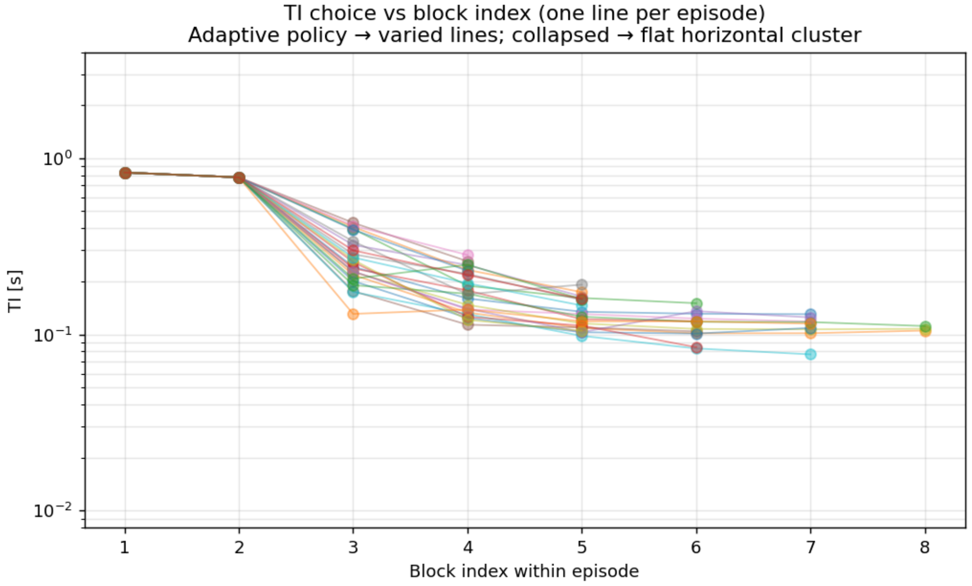


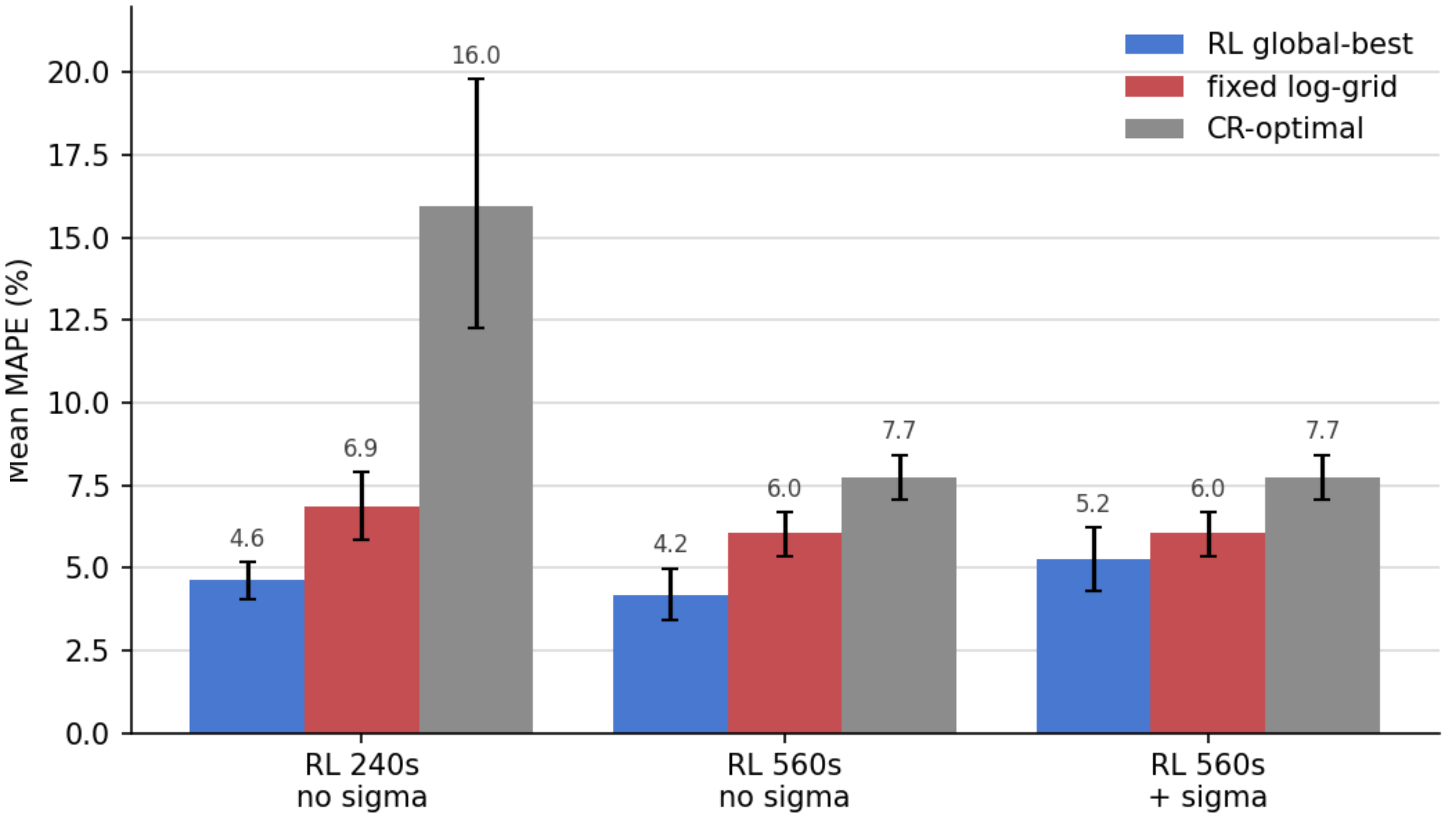
Run B: adaptivity helps on fewer spheres

5 spheres, T1 resampled every episode across full clinical range

Budget: 240s and 560s

σ: estimate of fitter uncertainity





(A3) Including previous TI values significantly reduces MAPE

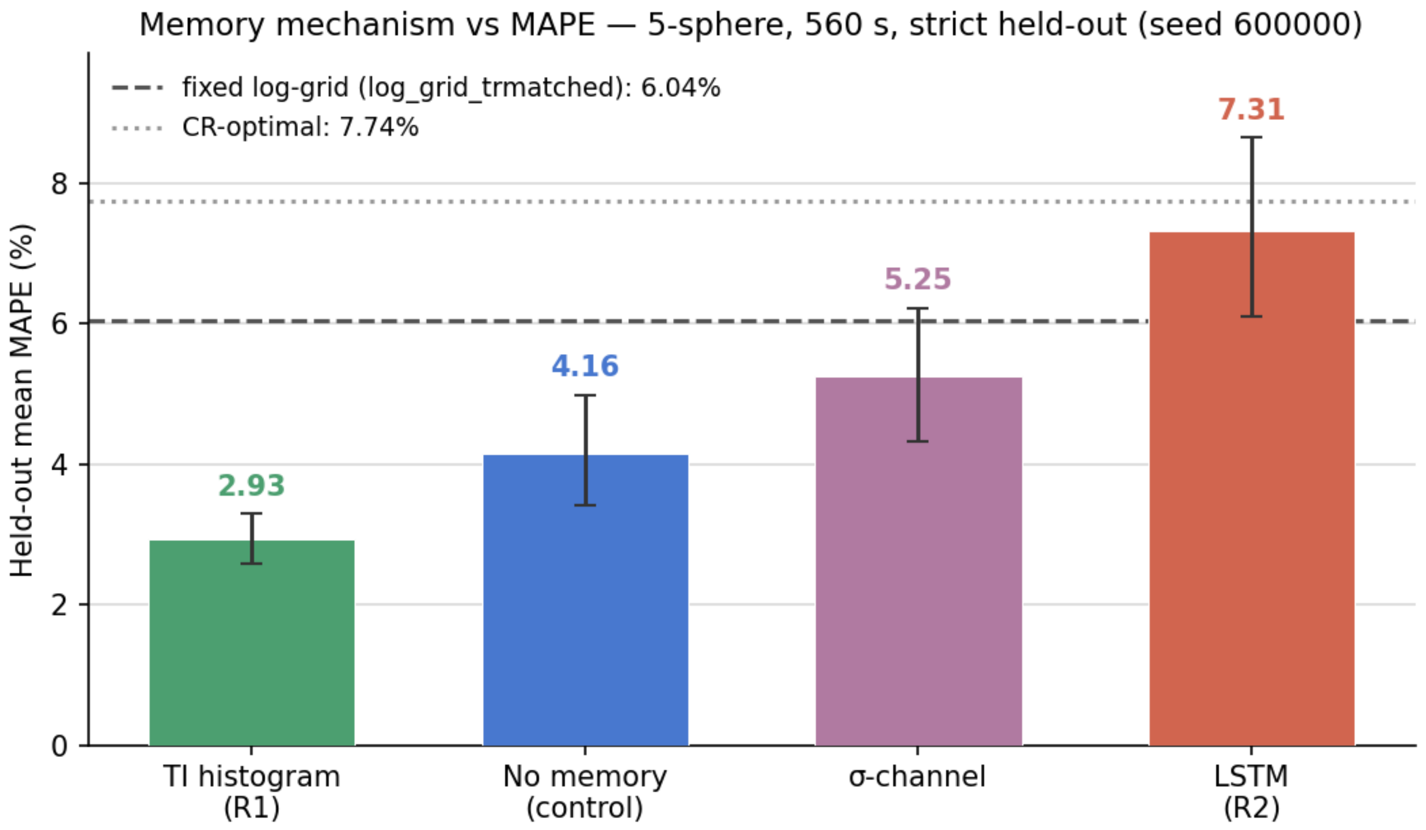
Agent state was augmented:

- With a histogram of previous TI acquisitions
- The per-sphere fitter uncertainity

An LSTM was used so the agent could learn a better representation of historical exploration

Best: 2.93% MAPE vs 6.04% (95.8% < 5%)

LSTM caveat: 5x slower step so undertrained

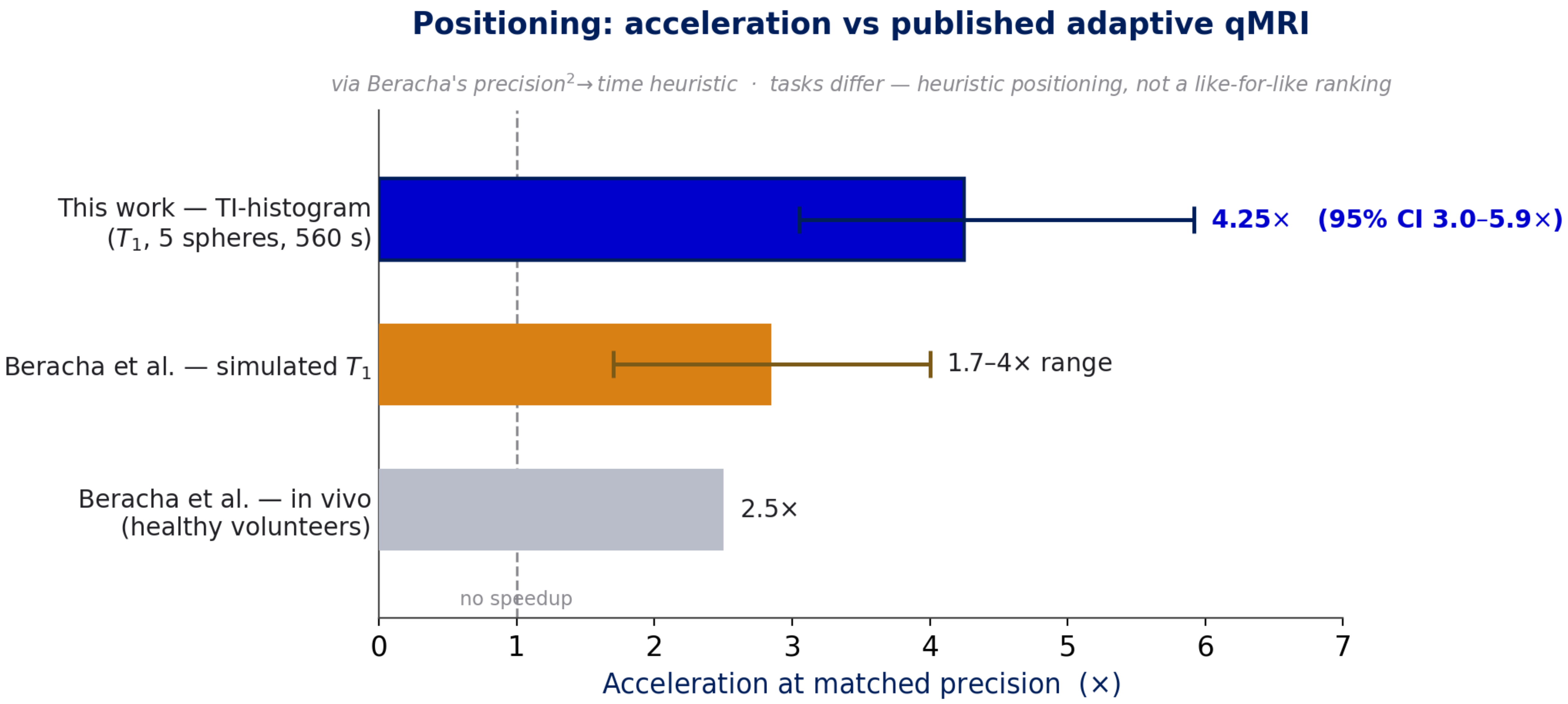


Future work & Limitations

- **Test T2 and PD plates (including parallel estimation using different sequences)**
- **Increase realism: use a slice selective RF pulse to isolate the 2D plane. Currently all spins outside the plane are dropped.**
- **Test other RL policies – use a replay buffer to reduce samples needed**
- **Real hardware validation**

Contributions & future work

- **Open source phantom twin: MRISystemPhantom.jl**
- **Validation of KomaMRI use a qMRI pipeline (39 -> 0.5%)**
- **First RL agent for adaptive qMRI**
- **Substantially reduced MAPE against fixed baselines (6 -> 2.9%)**



Positioning: acceleration vs published adaptive qMRI
via Beracha's precision²→time heuristic · tasks differ — heuristic positioning, not a like-for-like ranking
This work — TI-histogram
(T_1, 5 spheres, 560 s)
4.25× (95% CI 3.0–5.9×)
Beracha et al. — simulated T_1
1.7–4× range
Beracha et al. — in vivo
(healthy volunteers)
2.5×
no speedup
0
1
2
3
4
5
6
7
Acceleration at matched precision (×)

Submitting to ISMRT's annual international conference (Vancouver 2027)

INTERNATIONAL SOCIETY FOR MR

ISMRT

RADIOGRAPHERS & TECHNOLOGISTS

A Section of the ISMRM

Backup slides — anticipating Q&A

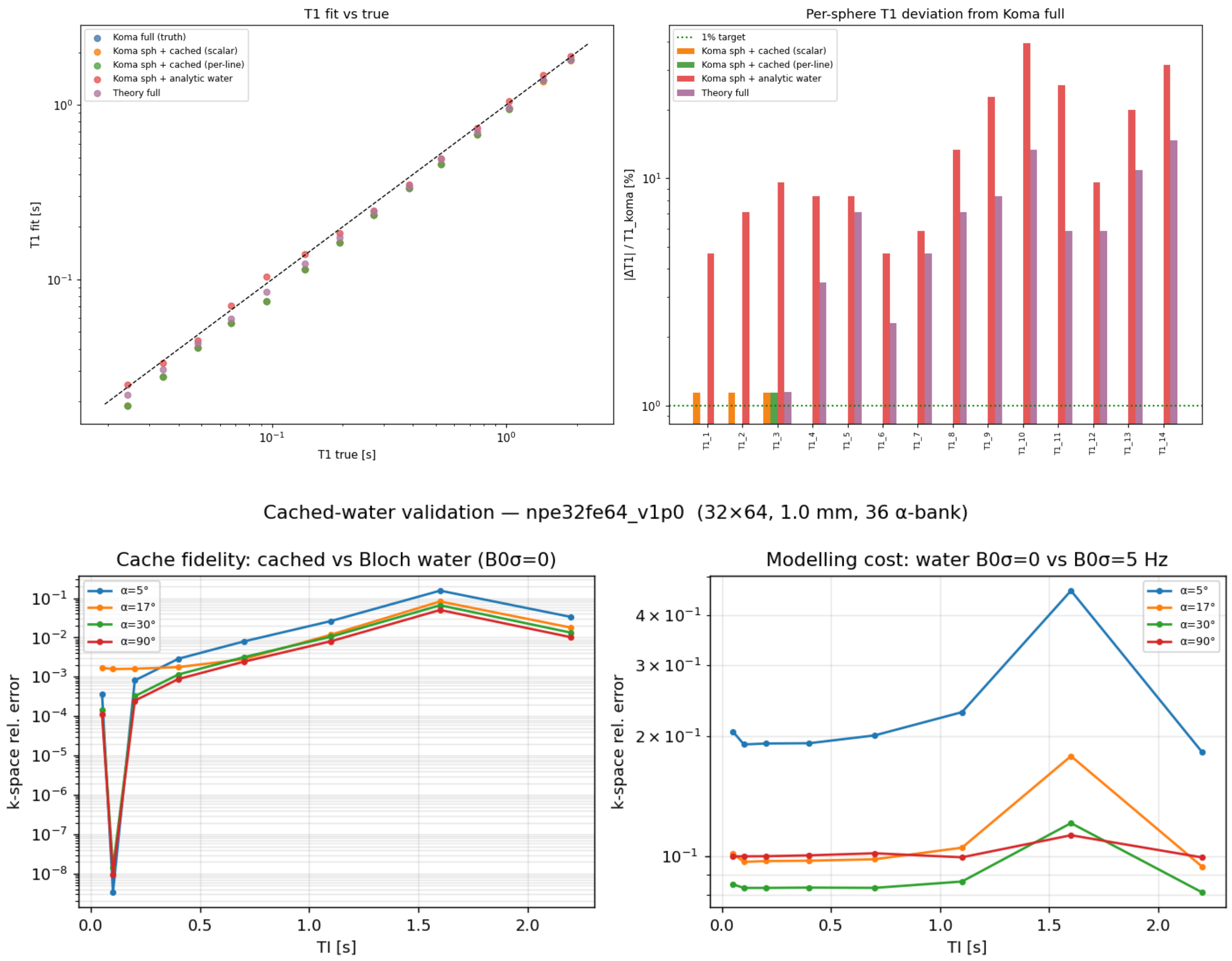
Backup — Reward screening (why Δlog MAPE)

Reward	Aggregation	30k PPO final MAPE
$\Delta \log \text{MAPE}$	mean	**0.830**
ΔMAPE	mean	0.839
terminal only	mean	0.896
ΔMAPE	$0.5\,\text{mean} + 0.5\,\text{max}$	0.866
$\Delta \log \text{MAPE}$	$0.5\,\text{mean} + 0.5\,\text{max}$	0.942
$-$MAPE / $-\log \text{MAPE}$	mean	1.000 (no blocks)

Table 5.2: Reward-screening study on the analytic RL environment (14-sphere T1 plate, 30k-step PPO screen). Progress rewards were necessary — the summed per-step penalties collapsed to a zero-measurement policy — and mean $\Delta \log \text{MAPE}$ gave the best learned policy. Worst-sphere weighting ($\alpha < 1$) did not help in the cheap screen.

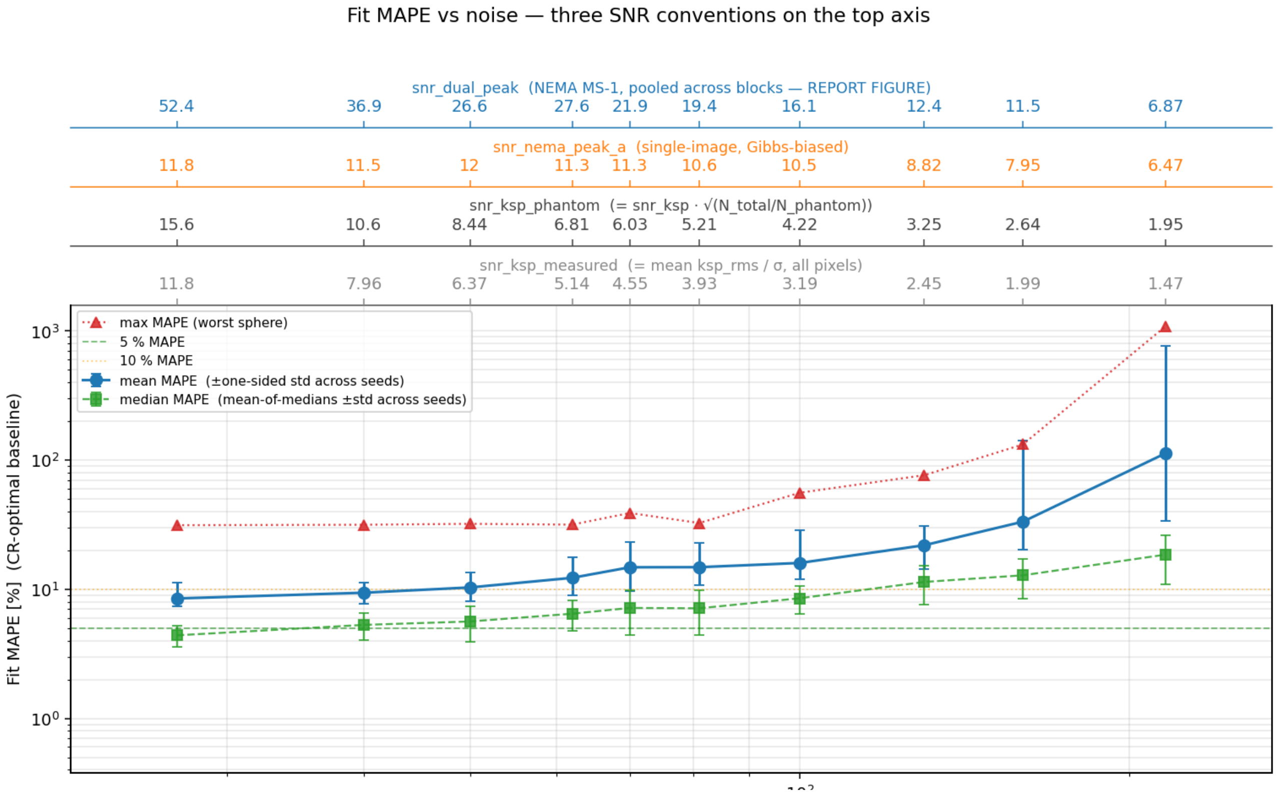
Backup — Cached-water validation

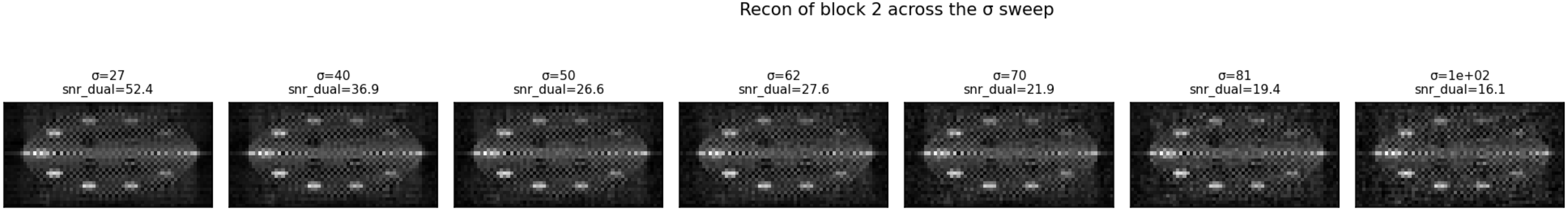
- Cached model matches full-Bloch T1 fits to 0.12% (worst 1.15%)
- Modelling water analytically instead is too crude: up to ~40% per-sphere deviation
- α-matched template bank: exact at matched flip angle, blended off-grid



Backup — Noise calibration (σ = 50)

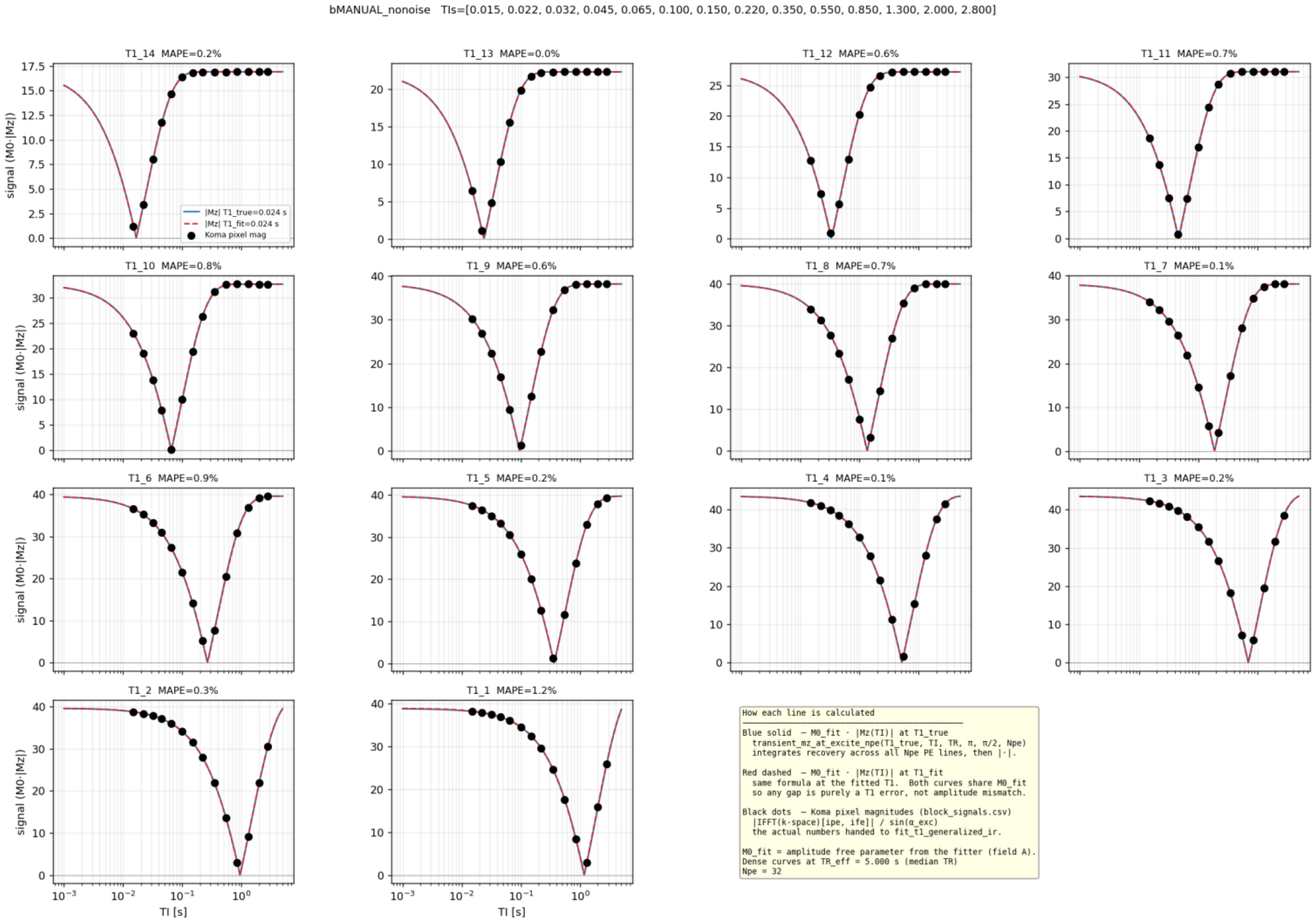
- σ=50 → NEMA SNR ≈ 17 (plate avg) to 28 (bright spheres) — clinical IR range
- 25-seed sweep: T1 fits degrade gracefully to σ≈85, collapse beyond





Backup — The second KomaMRI bug + fitter floor

- Bug 2: closing-knot collapse AFTER time-rebasing (T0 + t) — same ε mechanism, later in pipeline
  - reproducer: +52% jump at shot 18 (270 s), then a wrong plateau
- Fitter grid floor: 300 log-spaced T1 candidates → ~1% quantisation floor by construction
  - so reported MAPE can't fall below ~1%; 500-point grid reaches ~0.48%

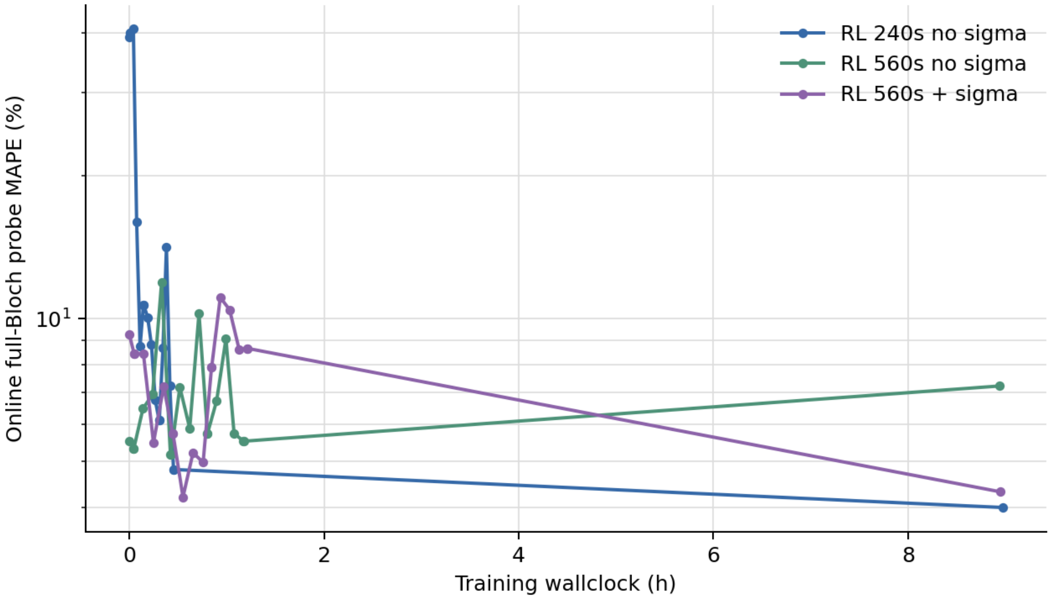


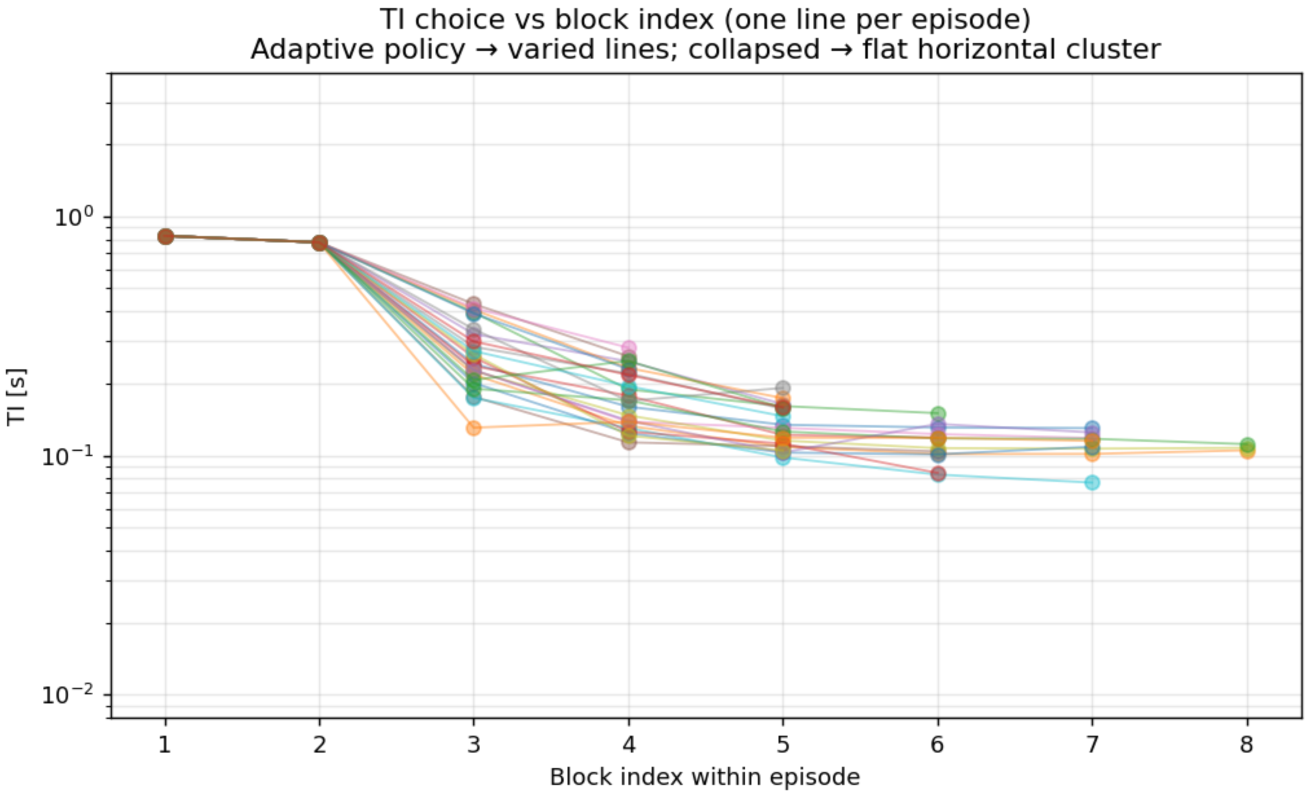
Backup — Run B policy behaviour

- 240 s policy: 5.92 blocks avg
- Zero timing repairs

- Opens with two high-TI probes (~0.78–0.83 s), then shorter adaptive refinements

- TI vs current estimate r=−0.72: a two-phase probe-then-refine, not a monotonic rule





Backup — Fixed comparators (Cramér–Rao baselines)

- Fixed schedules aren't hand-chosen — they're Cramér–Rao-optimal
- CRLB = variance floor for any unbiased estimator, set by design + noise
- static-OPTIMAL (solved for nominal fleet) vs static-AVERAGE (robust log grid)
  - CR wins when the fleet matches (Run A); collapses under per-episode variation (Run B)